

Assignment 2 -

The language of Business Analytics

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SECTION 1 - Foundational Concepts of Analytics

1. Business Analytics

Definition → Is the use of data, information, statistics and analytics methods in order to understand how a business is performing. It supports the decision-making in an organization. It usually involves the collection and interpretation of data to help a business identify trends, helping them make products or services that meets the demands of the customer (Evans, 2020)

Example: Shoprite ~~and~~ business analytics to analyse the sales of bread to see which brands sells the most so that they can decide how much stock to order.

Reference:

Citation Business Analytic (Evan 2020)

2. Data analytics

Definition: Data analytics is the process whereby raw unfiltered data is examined in order to find useful information and trends that companies can use to help make decisions. The objective of data analytics is to answer questions about possible outcomes (Evans 2020)

Example: Clicks shop analytics analysis which products do customers who buys together (buying pampers nappies together with baby wipes)

Citation: (Evan, 2020)

03: Business Intelligence (BI)

Definition: This is a set of processes and technologies that are used to convert raw data into ~~use~~ useful information that can be used to initiate increasing profits in a business.

Example: Samsung research teams are deciding on which features to include in the new S27 so that they can attract more customers to buy the new Smartphone.

Citation: Suresh Kashyap, business analytic

04: Data-driven Decision Making

Definition: It is the process of using data analysis in order to guide business decision making. It then enables the organisation to make smart informed, objectives and evidence based choices. This helps managers understand the situation before choosing which actions to take.

Example: Licks analyse data to find which sunscreen product sells the most during summer and decide to increase the stock in the store before it becomes in demand.

Citation: (Onesi-Ozigagum et al., 2024) (Arara et al., 2024)

05. Big Data

Definition

This refers to the evolution and use of technologies providing users with relevant information that is coming from large and complex amounts of information from many different sources that has been growing rapidly for a long time.

Example: Shoprite and Spar analytics analyses the typical family groceries purchases both in physical stores and online so that the company can decide which products to put on sale to attract more shoppers every month.

Citation: Riahi and Riahi, 2018

06. Descriptive Analytics

A type of data analytics that involves examining current or past data in order to understand past events, trends and patterns. Analytics use this to gain insight and understand what happened in the past. (Wolniak, 2023)

Example: Makro look at last months fridge sales and report that more fridges were sold as compared to this month because it was Black Friday.

Citation: (Wolniak, 2023)

07. Diagnostic analysis

Definition: Another branch of data analytics that focuses on past data in order to identify the cause of different outcomes, events or trends. This analysis is used to answer the questions of 'why did it happen?' (Wolniak, 2023)

examples: Takealot investigating why the sales of Samsung galaxy smartphones have decreased. The analysis investigate to check if the decrease in sales was because of price increase or competition

Citation (Evans, 2020)

08 Predictive analytics

Definition: ~~A~~ This type of analysis is used to analyze current and past data in order to predict what may happen in the future. This helps business prepare for what might happen in the future. (Kumar, 2018)

Examples: Shoprite using previous Christmas sales to predict how many meat products and soft drinks to stock up for this years festive seasons.

09. Prescriptive Analytics

Definition: Type of analysis that is focused on suggesting the best decision to be taken ~~based~~ in order to take advantage of the predicted future based on the available data. It is used to help businesses compare different choices and select the perfect solution. (Lepenioti et al., 2019)

Example: For the predicted stock at Shoprite for the next festive season, this analysis can be used to prescribe and recommend increasing stock and at different stores.

10. Data Literacy

Definition: It is defined as the ability to understand, collect, interpret and communicate information in a critical manner. This type of literacy involves the skills necessary to discover and access data, evaluate data quality as well as conducting analysis using data. (Downes, 2023)

Example: A manager at Clicks that is able to understand this type of literacy who is able to read the sales can use the information to decide what to and what not to order.

Citation: S Downes, 2023

Section 2- Data, Measurement and Reporting

11. Structured Data: This is data that is organised in a format or a layout, this type of data is composed in rows and columns making it easy to read and understand. It can also be defined as quantitative data and normally include data you can count and measure.

Example: Checkers employees database containing employees name, employees number, the department they belong to, the date they were hired, as well as their salaries

Citation: (Coursera, 2026)

12. Unstructure data

Definition: This is information that does not follow any formal format, structure or model. Unlike structured data, this type of data does not fit into rows or columns. This means that it is difficult to store, analyse and process. The data can include things such as images, videos, audio and text.

Example: Shein as a clothing company may collect information (unstructured data) from customer reviews, photographs taken as reviews and videos that is uploaded by customers to use it to find out why a particular clothing its sale is not increasing.

Citation: (Coursera, 2026)

13. Data Warehouse

Definition: This is where multiple database are stored and unlike on the database, here you can just store data but cannot manipulate it. In a data warehouse the information stored can be used by businesses for reporting and analysis. The warehouse is built to help support the analysis (Coursera, IBM)

Example: Mr Price retailer using a warehouse to store information about sales inventory, customer information from different systems into the data warehouse for analysis.

itation: (IBM, 2026)

14. Data mining

Definition: This is the process of going through a lot of data trying to find useful patterns, relationship or trends. The data mining is mostly in a form of computer-assisted technique used. It can change raw data into practical knowledge that business can use to solve challenges and analyze any potential future risk.

Example: Target data analytics using data mining can find that most customers who buy gaming laptops end up buying gaming mice and headphones together so they end up putting frequently bought together so that it can attract more customers.

Citation: (Amazon web Services, IBM, 2026)

15. Data visualisation

Definition: This is the process of visually presenting data using common graphics such as charts, graphs and other visual formats. The visualisation of such information makes it easy for large amount of data to be understood and can be noticed easily.

Example: Spar data analytics can use data visualisation to create graphs that show the sales of bread, milk, eggs and Cereal in different stores to check which store has higher sales.

Citation: Google cloud, IBM, Coursera, 2026)

16. Dashboard

Definition: This can be defined as a visual presentation or a screen that is showing important data points. The data points are put into one single view allowing the organizations to be able to monitor the business performance, identify problems early and make decisions without having to look at raw data.

Example: Using a dashboard to show the daily orders, stock, sales and revenue of different products like washing machines and dishwashers at Takealot.

Citation: Google Cloud, IBM

17. Metric

~~**Definition:** This is a measurement of~~

This is a measure of what is happening. It is used to track activities, monitor processes or the results of operations. The metrics provides visibility into operations of the business and its performance but it does not always directed to the success of the company.

Example: The number of people who visits Amazon and Takealot online can be measured and documented but since they are not customers they remain just a number of metrics of how many people visited the website.

Citation: Google Analytics, datapad. accessed 2026)

18. Key Performance Indicator (KPI)

Definition: This is a measurement of the company performance and its progress. The measure of success is based on the company goals and objective. The KPI in a company is necessary because it helps the organisation see if the company is making the right decisions or not.

Compared to Metrics the KPI is a metric used to measure success based on the company set goals, however a ~~metric~~ metric only measure the performance of certain business on an operational level.

Example: Takealot and Amazon ~~may~~ having an KPI of conversion Rate which may include the percentage of people who complete a purchase online instead of just visiting the website (Metric)

Citation: Datapad, Google Analytics

19. Benchmark

Definition: This is the process used by a business to compare the performance of the business against the industry standards, the company past performance and also from other business competitors.

Example: Samsung compares its sales satisfaction based on iPhone to see if most people prefer Samsung or Android.

Citation: Evans, J.R (2020).

20. Segmentation

Definition: It is the process of isolating and dividing big groups of customers into different groups mainly ~~becc~~ comprising of customers who have ~~shared~~ similar characteristics. The grouping can be based on behaviour, demographics. This process helps ~~customer~~ businesses to understand its customers more.

Example: Amazon separating and grouping customers based on their address for easy

Citation: (Kotler et al.; 2016)

SECTION 3- Customer and Sales KPIs

21. Conversion rate

Definition: This is the percentage of people who actually do what you wanted them to do out of everyone who held a chance

Example: If 1000 people visit a website and only 50 people end up buying it means that $\frac{50}{1000} = 5\%$

5% is the conversion rate.

Citation (Google Analytics, 2026)

22. Customer Acquisition Cost (CAC)

Definition: This is a measure of how much it cost to win one new customer, involving marketing and sales spend.

Example: Company like Mobicells spending R1000 on ads and only get 100 new customers, the CAC is R100 per customer

Citation: Investopedia (2026)

23. Customer Lifetime value (CLV/LTV)

Definition:

This is the total profit a business expects to make from one customer over the whole time they stick around.

Example: Netflix figures the average customer stays 3 years bringing a total of R3600 in profit

Citation: (Investopedia, 2026)

24. Churn Rate

Definition: This is the percentage of customers who stops using a certain product or stop doing business with you over a set of period.

Example: A gym with 200 members during the start of the year but loses 10 in a month after January.

Citation: (Investopedia, 2026)

25. Customer Retention Rate

Definition: This is the percentage of customers you keep over a period of time. Unlike the Churn Rate the customer retention rate measures the customers who remains loyal to a business.

Example: A gym monitoring the number of people who remains in the gym even after some cancels their membership during the start of the year.

Citation: (Investopedia, 2026)

26. Net Promoter Score

Definition: A score based on asking customers how likely they are to recommend your business to others.

Example: A hotel group gives surveys to guest.

Citation: (Qualtrics, 2026)

27. Average order Value (AOV)

Definition: This is the amount customers spend everytime they place an order. Calculated by $\frac{\text{total revenue}}{\text{orders}}$

Example: If Takealot receives a total revenue of R50000 from 1000 orders then the AOV will be R500

Citation: Hubspot Academy, Shopify (2026)

28. Sales funnel

Definition: These are the stages a potential customer passes through before becoming a loyal or staying customer.

Example: Seeing a Mado ad online for a TV, then putting the TV on a cart until you eventually decide to pay.

Citation: (Hubspot Academy, 2026)

SECTION 4 - MARKETING & CAMPAIGN ANALYTICS

29. Marketing campaign

Definition This is a planned set of marketing activities, the activities could be like ads, social media posts, working together towards one goal over a set of period

Example - A christmas ~~campaign~~ sales from Amazon running across TV, social media (facebook, Insta and TikTok)

Citation (Kotler et al., 2016)

30. Post-Campaign Analysis

Definition This is when we sit down and review the results of a marketing campaign after it ends. We do this to see what worked and what did not. We usually use metrics like ROI

Example - After the Christmas Campaign, Amazon can still sit down and the marketing team reports on which campaign brought the most sales.

Citation Hubspot Academy, 2026

31. Incremental Revenue

Definition: This is extra money that can be directed to a specific campaign.

Example: ~~If the sales were R5000 for a campaign~~
Shoprite normally expects R150 000 for the sales of cooking oil during the festive season but can generate R200 000 during a promotion.

Citation: McKinsey and Company

32. Return on Investment (ROI)

Definition: This is how much profit a business made compared to what was spent.

Example: Spending R1000 on ads that generate R4000, the profit made would be R3000 (ROI)

Citation: Investopedia

33. Return on Ad spend (ROAS)

Definition

This is how much you earn for every money spent on ads

Example: Pep spending R500 on ads that generates R2500 in sales, the ROAS is 5:1

Citation (Google Ads Help, 2026)

34. Click through Rate (CTR)

Definition: This is the percentage of people who saw the ad/link and those who actually clicked it.

Example: If 50 000 sees people sees an ad at takealot online but only 1000 people clicks on it, this means that ~~only~~ the CTR is 2%.

Citation: (Google Ads Help)

35. Bounce Rate

Definition: The percentage of people who on my page and often leave without anything

Example: Visiting Clicks online but leaves ~~before~~ without becoming meaningfully engaged.

Citation: (Google Analytics, 2026)

36. AB testing

Definition: This is the process of trying two version of something on different groups to see which one works better

Example: Takealot showing add of 20% off to one group and a R500 off to another group to see which one will produce more purchases

Citation: (Google Analytics, 2026)

37. Lead generation

Definition: This is the process of locating and collecting information from people so that they can potentially become customers, the information includes names, email and addresses

~~Example~~

Examples: Collecting information of businesses that are interested in purchasing furniture at Makro

Citation: Hubspot Academy

38. Marketing Attribution

Definition: This is the process of assigning credit to marketing touch points that resulted in the customer purchasing a product

Example: Seeing a Takealot product on Facebook and later checking on google and finally purchasing.

Citation: (Google Analytics Help)

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